Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building **cloud services**, **REST APIs**, and scalable data systems using **Node.js**, **Java**, Python, AWS, Docker, Kubernetes, SQL, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Built cloud data integrations using FastAPI, Supabase, SQL Server, and storage services, processing 5K+ monthly jobs with incremental loads, reconciliation, error handling, and **CI/CD**, achieving 99%+ reliability.
- Restructured fault-tolerant **AI workflows** using asynchronous workers, processing 1K+ jobs with explicit state transitions and failure isolation while increasing throughput threefold under concurrent production workloads without cascading failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Implemented a production **AWS backend** serving 150+ instructors and 10K+ daily grading requests, using automated deployments and rubric workflows to reduce feedback turnaround 60% while maintaining stable peak-load performance.
- Designed concurrent **REST APIs** with Python multithreading, AWS RDS, and secure file ingestion, processing 1K+ files per batch through fault-tolerant execution and validation across customer-facing grading workflows.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *LA, California*

- Built a Python pipeline processing 100K+ trajectory logs, improving long-horizon planning quality 18% through reproducible data preparation, model training, simulation evaluation, automated testing, and detailed experiment analysis.
- Architected a spatial-temporal encoder integrating HD-map geometry and historical agent trajectories through multi-head attention, improving closed-loop planning stability 15% while validating service outputs across diverse autonomous-driving scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Optimized large-scale **API integrations** using OOP patterns, AWS Glue, and S3, improving response times 30%; led scalable ETL workflows that raised pipeline efficiency 68% and reduced technical debt 20%.
- Engineered cross-platform microservices with automated builds and 20+ **unit and integration tests** using Jenkins, PyTest, and Mockito, supporting reliable releases across three operating systems within Agile sprints.

Skills

Languages: JavaScript, TypeScript, Java, Python, SQL, JSON, C++, C#
Backend & Data: Node.js, FastAPI, Flask, Spring Boot, REST APIs, OOP, Microservices, PostgreSQL, MySQL, SQL Server, MongoDB
Cloud & Infrastructure: AWS, EC2, RDS, S3, Azure, GCP, Docker, Kubernetes, Authentication, Authorization
DevOps & Quality: CI/CD, Jenkins, GitHub Actions, Git, Unit Testing, Integration Testing, PyTest, Mockito, Monitoring, Service Health, Agile/Scrum
Systems: Generative AI, LLM Workflows, Data Pipelines, Distributed Systems, Data Modeling, Scalability, Performance Optimization

Projects

CloudBeat Relay | Node.js, REST, Docker, Kubernetes

- Built and tested an authenticated **Node.js connected-device API** with JSON validation, idempotent command handling, health metrics, mocked publishing, Docker packaging, and Kubernetes readiness, liveness, replica, and resource controls.

Linguistix | Python, AWS, Docker, CI/CD

- Developed and deployed a containerized **RESTful data-processing API** on AWS ECS/Fargate with CodePipeline, automated CI/CD releases, validation checks, 87% accuracy, and production-grade reliability across repeatable cloud deployments.

Chat2SQL Code | Java, TypeScript, Node.js, SQL, Docker

- Worked across a **Java** and TypeScript codebase with Docker, Maven, Node.js, and Yarn, supporting 10+ relational and NoSQL databases through schema exploration, query authoring, and cross-platform workflows.